

The empirical recurrence for OEIS A283695: recovering a conjecture that is not written down

Adrian Perez Fontelles
Independent researcher

2 September 2026

Abstract

OEIS A283695 records “Empirical recurrence of order 56”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 102$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 56, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 56 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture one cannot read

OEIS A283695 is “Number of $5 \times n$ 0..1 arrays with no 1 equal to more than two of its horizontal, diagonal and antidiagonal neighbors.”. It has offset 1 and begins

32, 768, 8732, 150497, 2530461, 39372084, 640543058, 10379232484, ...

The entry states:

Empirical recurrence of order 56 (see link above)

As of the “Last modified” line on the live entry (Mar 14 2017 16:20:19, revision 4) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 800$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 102$ of the 800, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 102$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 204$ exact terms of the model returns order 56 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{56} c_i a(n-i),$$

c_1	14	c_{15}	−1975197624	c_{29}	−114564120898	c_{43}	8167898929
c_2	40	c_{16}	−3233532914	c_{30}	−45135358023	c_{44}	12206802672
c_3	352	c_{17}	2974394538	c_{31}	−254015322358	c_{45}	336705236
c_4	−6711	c_{18}	15167393053	c_{32}	166729091467	c_{46}	9182911178
c_5	−12061	c_{19}	11310281738	c_{33}	−181669048741	c_{47}	−5386801646
c_6	62939	c_{20}	−33454467696	c_{34}	52240351412	c_{48}	3940438656
c_7	493600	c_{21}	−40844046337	c_{35}	15348837094	c_{49}	−1427689992
c_8	670805	c_{22}	22358570076	c_{36}	−153081690411	c_{50}	526046640
c_9	−5102110	c_{23}	98805990855	c_{37}	225119511530	c_{51}	−58427736
c_{10}	−17272710	c_{24}	−39471930126	c_{38}	−3149661731	c_{52}	−25813880
c_{11}	−9747577	c_{25}	−124166492514	c_{39}	324310254108	c_{53}	−478544
c_{12}	133574311	c_{26}	−56783770834	c_{40}	−170976364256	c_{54}	241132
c_{13}	343167413	c_{27}	204734471044	c_{41}	13094591036	c_{55}	29276
c_{14}	−47094527	c_{28}	199247122314	c_{42}	−135573466986	c_{56}	1360

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 56, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $56 + 4$ terms removed returns order 56 as well, so $\deg q_{\min} = 56 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 16 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $56 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 56 that this sequence satisfies — and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A283695.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.